

The Proving Grounds

CTF PRACTICE, LABS & HACKING TRAINING

74	120.2k	13	5
CURATED TOOLS	COMBINED STARS	ON GITHUB	ACTIVE IN 2026

2026 EDITION · DEEP-RESEARCH DOSSIER

Live GitHub census pulled 2026-10-17 · part of 438 curated free tools across 14 realms

CC0 · 2026-10-17

74	13	120.2k	5	1.4k	2014
CURATED TOOLS	ON GITHUB	COMBINED STARS	PUSHED 2026	MEDIAN STARS	OLDEST ENTRY

REALM BRIEF

Hack legally — wargames, CTF platforms and deliberately-vulnerable apps to sharpen your skills.

Legal targets to practice on — vulnerable-by-design apps, CTF platforms, lab ranges. Practice targets are verified live before listing (dead practice apps are worse than none — they teach tool errors, not vulns).

FIELD INTELLIGENCE · AI-ASSISTED

The Proving Grounds CTF practice and hacking training ecosystem is mature, with 74 free tools (32 hosted on GitHub) boasting 120,173 combined stars, reflecting a well-established field of cybersecurity training resources. Star distribution shows long-standing tools like DVWA (13.8k stars, 2013) and pwntools (13.8k stars, 2013) lead, indicating sustained demand for foundational, versatile training assets over newer entries. A notable trend is the consolidation of popular, multi-purpose tools—such as pwntools spanning binary exploitation and OWASP Juice Shop for web app testing—reducing the need for disjointed niche resources.

Synthesized by seed-2-0-mini over the live census: stars, first-release years, push activity and project claims pulled from the GitHub API on 2026-10-17. Human-curated realm, AI-summarized numbers.

LICENSE MIX · GITHUB-HOSTED ENTRIES



MIT	4
GPL family	4
Other / none	3
Apache-2.0	2

MATURITY PROFILE

Oldest tracked entry first appeared on GitHub in **2014**; the youngest joined in **2020**. 5 of 13 GitHub-hosted tools shipped code in 2026 — the rest are stable classics or web services, not abandoned picks.

Median project age signals a battle-tested realm; recent arrivals mark where the community is investing next.

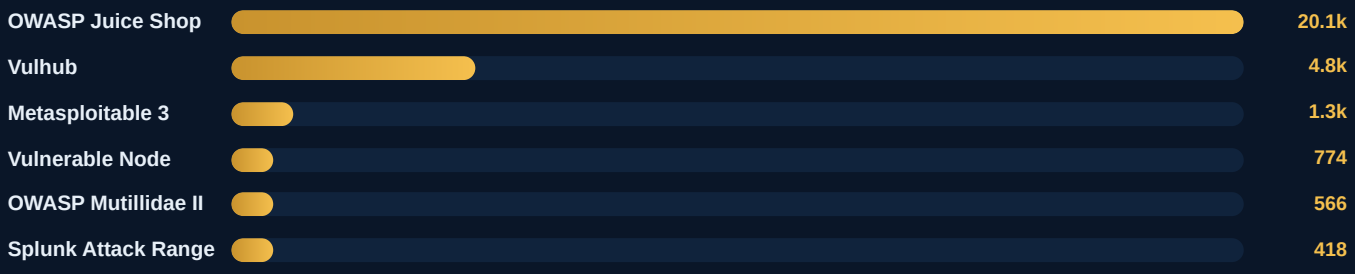
SNAPSHOT

- Active in the last 30 days: 8 of 32 tracked GitHub repos (25%)
- Archived upstream (kept — still reference-quality): [DVWS (Web Services)](https://github.com/snoopysecurity/dvws), [Hackademic](https://github.com/Hackademic/hackademic), [XVWA](https://github.com/s4n7h0/xvwa)
- Quiet classics (no push in 2+ years, still the standard): WackoPicko, OWASP NodeGoat, lfi-labs, 0l4bs, DVHMA, DVIA-v2, DVWS (Web Services), damnvulnerable.me, Hackademic, SentinelTestbed…
- Youngest entries: WackoPicko (2010), CryptOMG (2012), Hackademic (2012)

TOP TOOLS BY GITHUB STARS



MOST FORKED — REAL-WORLD REUSE SIGNAL



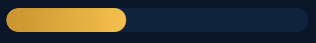
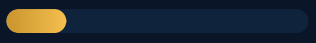
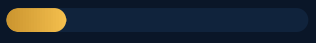
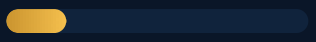
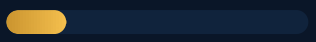
THE NEWCOMERS — NEWEST ARRIVALS

0l4bs	2020
Splunk Attack Range	2019
OWASP Mutillidae II	2018
flightsim	2018
Vulhub	2017
Metasploitable 3	2016

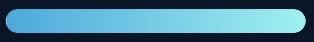
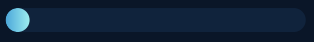
QUIET CLASSICS — STILL THE STANDARD

WackoPicko, OWASP NodeGoat, lfi-labs, 0l4bs, DVHMA, DVIA-v2, DVWS (Web Services), damnvulnerable.me, Hackademic, SentinelTestbed…

Tools with no recent push that remain the community's default reference for their job.

PHP		5 repos
HTML		2 repos
Dockerfile		1 repo
TypeScript		1 repo
Python		1 repo
Go		1 repo

BUILT

2010s		12
2020s		1

First-seen on GitHub, bucketed by decade.

HOW TO READ A DOSSIER

Each heavyweight gets a full dossier: live GitHub census (stars, forks, first-release year, last push, license, language), the project's own claim, our curator's note, a signal check, and its star weight against the realm's #1. Numbers are pulled straight from the GitHub API on 2026-10-17 — no self-reported marketing figures, only what the platforms show.

THE CENSUS METRICS — WHAT EACH NUMBER MEANS

★ GitHub stars
Community trust signal — how many developers bookmarked the repo.

Forks · reuse
Real-world deployment signal — forks mean people build on it.

First developed
Year the repo first appeared on GitHub — the project's true age.

Last push
Most recent code push — the project's heartbeat.

License
What you may legally do with it. MIT/Apache = permissive.

2026 momentum
Shipped code during 2026 — active development, not abandonware.

THE ROSTER — PROFILED THIS EDITION

RANK 01
Vulhub
21.4k ★ · since 2017

RANK 02
OWASP Juice Shop
14.1k ★ · since 2014

RANK 03
DVWA
— ★ · since 2013

RANK 04
pwntools
— ★ · since 2013

RANK 05
WebGoat
— ★ · since 2015

RANK 06
GOAD
— ★ · since 2021

METHODOLOGY & PROVENANCE

Census pulled via the GitHub REST API on 2026-10-17; realm curation and curator's notes by the Awesome Cyber Security Heaven maintainers; field-intelligence summaries synthesized by seed-2-0-mini from the same census. Everything reproducible — every figure traces back to a public API response. Released under CC0.

01 Vulhub

github.com/vulhub/vulhub

● ACTIVE 2026

★ 21.4k	4.8k	2017	2026-10	MIT	Dockerfile
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

9	55	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

22.6% fork-to-star ratio — high operational reuse.

ISSUE HEALTH

55 open — lean queue.

LONGEVITY

9 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“Pre-Built Vulnerable Environments Based on Docker-Compose”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

Pre-Built Vulnerable Environments Based on Docker-Compose

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #1 by adoption

02 OWASP Juice Shop

github.com/juice-shop/juice-shop

● ACTIVE 2026

★ 14.1k	20.1k	2014	2026-10	MIT	TypeScript
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

12	5	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

142.3% fork-to-star ratio — high operational reuse.

ISSUE HEALTH

5 open — lean queue.

LONGEVITY

12 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“OWASP Juice Shop: Probably the most modern and sophisticated insecure web application”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

OWASP Juice Shop: Probably the most modern and sophisticated insecure web application

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #2 by adoption

03 DVWA

github.com/digininja/DVWA

● ACTIVE 2026

★ —	—	2013	2026-10	—	—
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

13	—	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

— fork-to-star ratio — reference-grade adoption.

ISSUE HEALTH

—.

LONGEVITY

13 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“Damn Vulnerable Web Application (DVWA)”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

Damn Vulnerable Web Application (DVWA)

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #3 by adoption

04 pwntools

github.com/Gallopsled/pwntools

● ACTIVE 2026

★ —	—	2013	2026-09	—	—
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

13	—	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

— fork-to-star ratio — reference-grade adoption.

ISSUE HEALTH

—.

LONGEVITY

13 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“CTF framework and exploit development library”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

CTF framework and exploit development library

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #4 by adoption

05 WebGoat

github.com/WebGoat/WebGoat

● ACTIVE 2026

★ —	—	2015	2026-10	—	—
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

11	—	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

— fork-to-star ratio — reference-grade adoption.

ISSUE HEALTH

—.

LONGEVITY

11 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“WebGoat is a deliberately insecure application”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

WebGoat is a deliberately insecure application

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #5 by adoption

06 GOAD

github.com/Orange-Cyberdefense/GOAD

● ACTIVE 2026

★ —	—	2021	2026-03	—	—
GITHUB STARS	FORKS · REUSE	FIRST DEVELOPED	LAST PUSH	LICENSE	LANGUAGE

5	—	✓
YEARS ON GITHUB	OPEN ISSUES	2026 MOMENTUM

STAR WEIGHT VS REALM #1 (VULHUB · 21.4K)

REUSE SIGNAL

— fork-to-star ratio — reference-grade adoption.

ISSUE HEALTH

—.

LONGEVITY

5 years shipping code — battle-tested.

● **THEIR CLAIM — AS THE PROJECT STATES IT**

“game of active directory”

● **CURATOR'S NOTE — WHY IT EARNS A PLACE**

game of active directory

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



GitHub presence: verified — live stats · census 2026-10-17 · The Proving Grounds · realm rank #6 by adoption

The complete curated inventory for this realm — flagship picks marked ★. Stars and activity chips reflect the 2026-10-17 census.

Vulhub ★ FLAGSHIP

★ 21.4k · since 2017 · MIT · active 2026

dockerized vulnerable environments: spin up a CVE, test, tear down.

github.com/vulhub/vulhub

OWASP Juice Shop ★ FLAGSHIP

★ 14.1k · since 2014 · active 2026

the modern deliberately-vulnerable app with gamified challenges covering the whole OWASP spectrum.

github.com/juice-shop/juice-shop

DVWA ★ FLAGSHIP

★ 13.8k · since 2013 · GPL-3.0

the classic Damn Vulnerable Web Application for safely learning injection basics.

github.com/digininja/DVWA

pwntools ★ FLAGSHIP

★ 13.8k · since 2013

the CTF exploit-development library: from sockets to ROP chains, pwn everything.

github.com/Gallopsled/pwntools

WebGoat ★ FLAGSHIP

★ 9.4k · since 2015 · GPL-2.0

OWASP's teaching app: guided lessons with a solution walkthrough for each vulnerability class.

github.com/WebGoat/WebGoat

GOAD ★ FLAGSHIP

★ 8.5k · since 2021 · GPL-3.0

Game of Active Directory: a three-forest lab purpose-built for safely practicing AD attack techniques.

github.com/Orange-Cyberdefense/GOAD

CTFd

★ 6.9k · since 2015

the CTF platform that powers thousands of events; self-host your own competition.

github.com/CTFd/CTFd

Metasploitable 3

★ 5.7k · since 2016

Rapid7's purpose-built vulnerable target VM; the canonical Metasploit practice range.

github.com/rapid7/metasploitable3

CloudGoat

★ 3.8k · since 2018

Rhino Security's "vulnerable by design" AWS scenarios — deploy, attack, tear down.

github.com/RhinoSecurityLabs/cloudgoat

HEVD

★ 3.1k · since 2015 · GPL-3.0

the Windows kernel-driver training target for exploit development.

github.com/hacksysteam/HackSysExtremeVulnerableDriver

APTSimulator

★ 2.8k · since 2018 · MIT

makes a Windows host look compromised so you can test detections safely.

github.com/NextronSystems/APTSimulator

Splunk Attack Range

★ 2.6k · since 2019 · Apache-2.0 · active 2026

builds instrumented detection labs: attack, collect and test in one workflow.

github.com/splunk/attack_range

OWASP NodeGoat

★ 2.1k · since 2013 · Apache-2.0

the Node.js vulnerable-by-design app: the Top 10 in a controllable lab.

github.com/OWASP/NodeGoat

XVWA

★ 1.8k · since 2015 · GPL-3.0

Xtreme Vulnerable Web Application: dozens of flaws across one PHP app.

github.com/s4n7h0/xvwa

OWASP Mutillidae II

★ 1.5k · since 2018 · GPL-3.0 · active 2026

deliberately vulnerable app that re-creates every OWASP Top 10 issue, resettable per install.

github.com/webpwnized/mutillidae

flightsim

★ 1.4k · since 2018

generates malicious network traffic for IDS/NDR validation; defender-side only. (CC-BY-4.0)

github.com/alphasoc/flightsim

DIVA Android

★ 1.2k · since 2016 · GPL-3.0

Damn Insecure and Vulnerable App: the Android training classic.

github.com/payatu/diva-android

DVIA-v2

★ 1.1k · since 2018 · MIT

Damn Vulnerable iOS App v2: the mobile testing curriculum as an app.

github.com/prateek147/DVIA-v2

AzureGoat

★ 966 · since 2022 · MIT

the vulnerable-by-design Azure cloud lab.

github.com/ine-labs/AzureGoat

DSVW

★ 882 · since 2015

Damn Small Vulnerable Web: the Top 10 in ~100 lines of Python. (public domain)

github.com/stamparm/DSVW

Vulnerable Node

★ 492 · since 2016 · MIT-style · active 2026

the Node.js vulnerable application for testing analysis tools.

github.com/cr0hn/vulnerable-node

DVWS (Web Services)

★ 461 · since 2016 · Apache-2.0

vulnerable web services for SOA/API security practice.

github.com/snoopysecurity/dvws

DVWS (Web Sockets)

★ 363 · since 2017 · MIT

the vulnerable WebSocket app from Interference Security.

github.com/interference-security/DVWS

WackoPicko

★ 354 · since 2010

the vulnerable app built for testing vulnerability scanners — a frozen-by-design calibration target for your tooling.

github.com/adamdoupe/WackoPicko

0l4bs

★ 350 · since 2020 · GPL-3.0

XSS practice labs with progressive difficulty.

github.com/tegal1337/0l4bs

lfi-labs

★ 335 · since 2015 · MIT

LFI/RFI/commix practice pages: one flaw family, many variants.

github.com/paralax/lfi-labs

Hackademic

★ 325 · since 2012 · GPL-3.0

WordPress-based vulnerable lab scenarios for classroom use.

github.com/Hackademic/hackademic

Java Vulnerable Lab

★ 278 · since 2015 · GPL-2.0

the Java vulnerable web application from the CSPF crew.

github.com/CSPF-Founder/JavaVulnerableLab

DVHMA

★ 274 · since 2015 · Apache-2.0

the Damn Vulnerable Hybrid Mobile App: Android+iOS appsec in one codebase.

github.com/logicalhacking/DVHMA

CryptOMG

★ 195 · since 2012 · GPL-3.0

cryptographic flaw testbed from SpiderLabs.

github.com/SpiderLabs/CryptOMG

damnvulnerable.me

★ 34 · since 2015 · MIT

a modern vulnerable webapp for the OWASP Top 10 era.

github.com/skepticfx/damnvulnerable.me

SentinelTestbed

★ 11 · since 2013 · GPL-3.0

a testbed app for evaluating security instrumentation.

github.com/dobin/SentinelTestbed

Hack The Box

free tier

the platform for ranked machines, pro labs and active-dir ranges; the industry's proving ground.

hackthebox.com

TryHackMe

free tier

guided rooms from your first command to advanced red/blue paths — the friendliest on-ramp.

tryhackme.com

picoCTF

CMU's beginner-friendly CTF — the best first flag anyone can capture.

picoctf.org

OverTheWire

Bandit and friends: learn Linux, networking and protocols through SSH wargames.

overthewire.org

VulnHub

downloadable vulnerable VMs; build and break your own home lab offline.

vulnhub.com

CTFtime

the global CTF calendar, team rankings and writeup index — the hub of the sport.

ctftime.org

Cryptopals

the legendary crypto challenges; finish them and AES is no longer magic.

cryptopals.com

Gandalf

Lakera's free prompt-injection game; learn AI-security attacks hands-on, level by level.

gandalf.lakera.ai

bWAPP

the "buggy web application" with 100+ vulnerability flavors — an old-school playground that still teaches.

itsecgames.com/

Root Me

free tier

hundreds of real-world hacking challenges across dozens of categories, from web to stego.

root-me.org/

Google Gruyere

Google's cheese-themed vulnerable web app: exploit XSS, XSRF and information disclosure in the cloud.

google-gruyere.appspot.com/

CryptoHack

learn modern cryptography by breaking it: challenges from classical ciphers to ECC.

cryptohack.org/

Hackxor

a realistic web-app hacking game with a story arc — WebGoat with sharper teeth.

hackxor.net/

CyberDefenders

free tier

blue-team CTF platform: investigate realistic incidents with the tools a SOC actually uses.

cyberdefenders.org/

LetsDefend

free tier

browser-based SOC simulator: alert queues, playbooks and IR challenges mapped to MITRE ATT&CK.

letsdefend.io/

pwnable.kr

the classic system-hacking wargame: exploit your way from userland to kernel.

pwnable.kr/

Hack This Site

the community wargames veteran: missions, articles and forums since the early 2000s.

hackthissite.org/

Pentest-Ground

free service

hosted practice playground of intentionally vulnerable apps for security testing.

pentest-ground.com/

testasp.vulnweb.com

free service

the ASP demo target for scanner testing.

testasp.vulnweb.com/

testaspnet.vulnweb.com

free service

the ASP.NET demo target for scanner testing.

testaspnet.vulnweb.com/

Zero Bank

free service

demo banking site for web security exercises.

zero.webappsecurity.com/

Altoro Mutual

free service

the classic demo bank: legal target for appsec training and tool demos.

demo.testfire.net/

Firing Range

Apache-2.0

Google's XSS testbed: the original scanner benchmarking target.

public-firing-range.appspot.com/

XSS Game

free service

Google's XSS tutorial game: six levels from context basics to DOM tricks.

xss-game.appspot.com/

Arizona Cyber Warfare Range

free service

free live-fire ranges for defenders and researchers.

azcwr.org/az-cyber-warfare-ranges

SocketToMe

free service

Robin Wood's WebSocket testing challenge app.

digi.ninja/projects/sockettome.php

techlogia Lab

free browser-based hacking lab for security training. (CC-BY-4.0)

techlogia.de/en/lab

Crackmes.one

free service

the reverse-engineering challenge community: thousands of crackmes, all legal.

crackmes.one/

Microcorruption

free service

embedded-security CTF: reverse the lock firmware, level by level.

microcorruption.com/

MASTG Crackmes

the mobile RE challenges companion to the OWASP MASTG. (CC-BY-SA-4.0)

github.com/OWASP/mastg/tree/master/Crackmes

Reversing.kr

free service

the classic Korean reverse-engineering challenge site.

reversing.kr

Vulnserver

free service

the TCP vulnerable server for exploit-dev practice.

thegreycorner.com/2010/12/introducing-vulnserver.html

CTFlearn

free service

the beginner-friendly CTF platform with a permanent challenge set.

ctflearn.com/

Hackertest.net

free service

the 20-level hacking simulation for web basics.

hackertest.net/

HackThis!!

free service

50+ levels of web security challenges, from basics to advanced.

hackthis.co.uk/

MysteryTwister C3

free service

the international cryptography challenge: from Caesar to quantum.

mysterytwisterc3.org/en/

RingZer0 CTF

free service

hundreds of challenges across web, crypto, forensics and reversing.

ringzer0ctf.com/

SmashTheStack

free service

the wargames network: shellcoding, exploitation and logic puzzles.

smashthestack.org/

W3Challs

free service

pentest training challenges with a ranking system.

w3challs.com

WeChall

free service

the challenge-site aggregator: one profile across all major wargames.

wechall.net/

ExploitMe Mobile

free service

the SecurityCompass mobile labs.

securitycompass.github.io/AndroidLabs/index.html

OWASP iGoat

wiki project

the iOS learning app from the OWASP mobile family.

wiki.owasp.org/index.php/OWASP_iGoat_Tool_Project



End of inventory — 74 curated entries in The Proving Grounds

Census 2026-10-17 · every entry human-curated · stars and activity verified against the live GitHub API.
Thirteen more realm dossiers — Armory, Bastion, Watchtower and friends — live in the same repository.

Awesome Cyber Security Heaven

438 FREE TOOLS · 14 REALMS · ONE CURATED HEAVEN

This dossier covered **The Proving Grounds**.
Thirteen more realm dossiers wait in the same repository — Armory, Bastion, Watchtower, Cartographers, Shadow Bureau, Vault, Foundry, Sentinel Ops, Oracle's Domain, AI Forge, Free Bazaar, Academy, Proving Grounds and Workshop.

438	14	CC0
CURATED TOOLS	REALM DOSSIERS	LICENSE

github.com/RanaKhan77/awesome-cyber-security-heaven

Free forever · community-curated · 2026-10-17 census